The Intuition
Variational Proof Structures

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Preface

The last volume's preface earned the number four: three soundings leave a point and its mirror twin, and the fourth, off-plane, breaks the tie. This preface asks the question a careful reader asked next, and answers it, because the answer is this volume's first lesson in how a representation behaves. Why stop at four? Why not five soundings, or six, or a hundred? Instruments are cheap; take *many*. The answer — and it deserves to be earned, not waved — is that many buys nothing, and the reason it buys nothing is a fact about representation itself.

Watch the fifth sounding arrive. Four have done their work: the candidates were cut to two, the tie was broken, the point has been named. Now the fifth reading comes in — and there is nothing left for it to say. If the apparatus is honest, the fifth reading must pass through the point the four already fixed; its value is implied before it is taken. It can *confirm* — and confirmation has real uses — but it cannot *inform*: no candidate remains for it to eliminate. And what holds for the fifth holds for the fiftieth. Once a representation with three coordinates and one tiebreak has been filled, every further constraint is spoken for in advance. The trade's word for this is overdetermination, and its classical mathematics is not folklore but a named subject: the method of least squares, published by Legendre in 1805 and given its foundation by Gauss — who had used it privately before the publication, and who built its theory in 1809 and again in 1821 — precisely for the overdetermined systems of astronomy and geodesy, where observations outnumber unknowns as a matter of course. The reader should note what the method *does* with the surplus: it does not extract new coordinates from it. It manages error. The honest job of the pile has been, since the year it was named, consistency and noise — never information. And since this page has now earned it, the number may as well receive its definition here, where it stands: *five is the first reading that gives no information* — the first constraint spoken for in advance, able to confirm what the filled representation has already named and unable to eliminate anything, because nothing remains to eliminate. Four is the count that completes the representation; five is the name of the surplus.

That distinction — noise against structure — is the earn's load-bearing beam, and the last volume already set it. Many measurements are the correct response to *noise*: readings scattered by accident average toward truth, and the pile genuinely helps. But the two-candidate ambiguity of trilateration was never noise; the last preface said so in so many words — it is structural, it lives in the geometry, not the noise — and against structure the pile is helpless. A thousand repetitions of the same three soundings polish the same two candidates to a brilliant shine and choose neither. What broke the tie was not more of the same; it was one measurement of a *different character*, placed off the plane. Structure yields to structure, never to volume. So the case for many collapses into two honest halves: against noise, many works and is welcome; against structure, many is spoken for in advance — and the naming of a point in a filled representation is structure through and through. The representation was complete at four, and completeness is indifferent to enthusiasm.

The reader who has followed this series will hear a familiar chord. The drift toward many — more soundings, more repetitions, more and finer forever — is the drift toward the continuum, and this series has held a fence against that drift since its first volume: the counts that matter have been small and structural throughout, and the pile has never once been where the content lived. This preface adds the surveyor's case to that standing discipline and claims nothing further from it.

And now the fence, in the same breath, because it is the load-bearing half of everything above. Nothing in this preface is a claim about the world — not about its dimensions, not about what space is, not about how many directions reality possesses. The four that sufficed and the many that is unnecessary are facts about a representation*: about how many independent constraints fill a description that carries three coordinates and one tiebreak. A reader could build a representation with more coordinates or fewer and the arithmetic would move with the representation, which is exactly the point — the count belongs to the description, and the world is left un-claimed. This series said on the first page of its first volume that the electron, in these books, is the name of a mathematical model; the same licence governs here, applied to dimensionality. Whoever reads the four of the last preface, or the completeness of this one, as this series measuring the dimensions of spacetime has taken a step no volume of this series takes.*

One more law belongs on this first page, because this volume is licensed to do something its predecessor forbade, and a licence without a law is a leak. The last volume permitted only cited ideas; this one is called *The Intuition*, and it will hand the reader pictures — the soap film, the mountain pass, the bookkeeper's zero, the two hands closing — because pictures are how proof structures are *felt* before they are checked. The law is this: every intuition in this book is an aid, and it is always marked as one — *intuition offered, not claim made* — and no intuition may add a single mathematical claim that its cited sources do not already contain. An intuition here is a way of seeing cited material, never new content; the back matter will keep a ledger of every intuition beside the citation it re-sees; and the failure this book hunts in itself is the intuition that quietly hardens into an assertion. Claims keep the last volume's law — a name and a date, or they do not appear. Pictures keep this one's — a label, or they do not appear.

With the number earned, the fence set, and the law posted, the volume can say in one sentence what it is for: four faces of mathematics — geometry, topology, algebra, and numerics — each read as a structure a variational proof can take, so that the constructions of the last volume and the measurements of the first stop being merely verified and start being obvious. Read on with the label in mind.

Chapter 1

The First Variation

> INTUITION OFFERED, NOT CLAIM MADE. *Dip a bent wire in soap solution and lift it out: the film that spans it is the least area the wire permits. Hang a chain from two nails: the curve it settles into is the one that puts its weight lowest. Watch light cross from air into water: of all the bent paths it might take, it takes the one that spends the least time. Three benches, one feeling — that nature, offered every candidate, somehow picks the best one — and the feeling is worth having before it is examined, because a whole face of mathematics is the examination.*

The examination began as a public dare. In June of 1696, Johann Bernoulli posed a challenge to "the sharpest mathematicians in the whole world": given two points, one above the other but not vertically so, find the curve down which a bead slides under gravity in the least time. The answer is not the straight line — the straight line is shortest but not fastest — and the problem, the *brachistochrone*, is the subject's founding legend for what happened next. Newton, by the account that has come down through his niece, returned home from the Mint at four in the afternoon, found the challenge, and had solved it by four the next morning; he published the solution anonymously, and Bernoulli, reading it, is said to have remarked that he knew the lion by his claw. The curve is the cycloid, the same curve traced by a point on a rolling wheel, and five solutions arrived in all — both Bernoullis, Leibniz, l'Hôpital, and the anonymous lion. What matters to this chapter is not the roster but the kind of question: not *what is the value of this quantity* but *which curve, among all candidates, is extreme* — and a question quantified over all candidates needs a new kind of proof.

The proof structure that answers it was built in two generations. Euler, in his treatise of 1744, gave the first systematic method and the equation that bears the subject's fingerprint: a candidate curve is extremal exactly when a certain differential condition holds along it. Then Lagrange — nineteen years old, writing to Euler in 1755 — replaced Euler's geometric derivation with the analytic device the subject still uses, the variation itself: perturb the candidate by an arbitrary small admissible wiggle, expand to first order, and demand that the first-order change vanish for *every* admissible wiggle. The attribution deserves its sentence, in this series' tradition: Euler, receiving the young man's method, abandoned his own, adopted Lagrange's, gave the subject the name it still carries — the calculus of variations — and said whose the method was. Lagrange's mature statement is the *Mécanique analytique* of 1788; Fermat's least time, which the soap-and-light intuition above leans on, was stated for optics in 1662 and stands as the structure's great anticipation.

Now say the structure plainly, because it is the chapter's cargo and the face's signature. To prove a candidate extremal, one does not compare it with each rival — the rivals are uncountable. One *varies* it: adds an arbitrary admissible perturbation, differentiates the quantity of interest along that perturbation, and shows the derivative vanishes at the candidate, whatever the perturbation was. The universal quantifier — *for every* wiggle — is carried not by exhaustion but by the arbitrariness of one symbol, and the vanishing condition condenses, by the Euler–Lagrange argument, into a differential equation checked pointwise along the candidate. That is the first variational proof structure: a claim about a whole ocean of candidates, proved by one derivative taken in a direction left arbitrary. The reader has owned its most important instance since the last volume: the geodesics of Chapter 7 there — the locally-shortest paths the ruler selects — are exactly such theorems. There, they were objects; here, the reader is asked to re-feel them as *proofs* — a geodesic is a statement whose demonstration is a vanishing first variation — and that re-feeling is what this volume means by its title.

> INTUITION OFFERED, NOT CLAIM MADE. *The feeling that carries the structure: stand on the candidate and jiggle. If every jiggle, to first order, costs nothing and gains nothing, you are standing somewhere special. The proof is the jiggling made honest — one arbitrary jiggle standing for all of them.*

And now the fence, because the opening feeling — *nature picks the best*

— contains a word this book must not let harden. “Best” is a property of the functional some person chose to extremize: least area for the film, least potential energy for the chain, least time for the ray. The mathematics proves that the observed configuration extremizes that chosen quantity; it does not prove that nature *prefers*, *intends*, or *deliberates*. The light ray does not survey its options; the variational statement and the pointwise differential law are two spellings of one content — Lagrange’s own treatise exists to translate between them — and the appearance of choice is the reader’s, imported with the word “best.” The intuition remains a fine way to find the functional and to feel the proof; teleology is what it must not become, and this paragraph is the label saying so.

What the first variation cannot see is the chapter’s honest limit, and the next chapter’s opening. A vanishing first derivative marks the flat places — but the mountain pass is as flat, at its saddle, as the valley floor. Which flat places are stable, which are passes, and what the answer has to do with the *shape* of the space the candidates live in — that question needs the second variation, and it carries this volume to its second face.

Chapter 2

The Second Variation

> INTUITION OFFERED, NOT CLAIM MADE. *Stand at a mountain pass. Underfoot it is as flat as any valley floor — a marble placed exactly there would rest — and yet the flatness lies. Step along the ridge and you fall away to either side; step across it and you climb. The valley floor and the pass agree completely at first order and differ in everything that matters, and the difference is invisible to any instrument that only asks whether the ground is level. The feeling to carry: flatness at first order hides the real story, and the real story is told by what happens at second order — and, deeper, by the shape of the country that made a pass necessary at all.*

The first half of the story is classical and carries two names. The last chapter's structure finds the flat places — first variation zero — and stops; to ask which flat places are minima, the probe must go one order deeper. Vary the candidate twice: the second variation is a quadratic reading of the wiggle, and its sign pattern classifies the flat place — positive against every wiggle, a floor; negative against every wiggle, a crest; mixed, a pass. Legendre, in 1786, gave the pointwise condition the sign demands of the integrand, and Jacobi, in 1837, found the deeper and stranger fact: follow a family of extremals out from a point, and there can come a definite station — the conjugate point — past which a path that minimized honestly stops minimizing, its second variation acquiring a direction of descent. The reader has met the physical shadow of this on a globe: the great-circle arc from a city is genuinely shortest — until it passes the antipode, its conjugate point, after which a shorter way around exists. The second variation is not a bookkeeping refinement; it is where the geometry of the *space of candidates* first speaks.

And that is the door to this chapter's real face. Ask the naive question: could the pass be gotten rid of? Flatten it, push it down, deform the landscape until only floors remain? For an isolated function on a featureless space, perhaps. But on a country with *shape* — and here the reader should picture walking on the surface of a doughnut — the answer is no, and the no is a theorem. Assign heights to the doughnut's surface any smooth way you please: there must be a lowest point and a highest, and there must *also* be passes — at least two of them — because the surface's two independent ways of looping cannot be filled in by any rearrangement of heights. Marston Morse, in the work gathered in his 1934 *The Calculus of Variations in the Large*, made this exact bargain precise: the critical points of a function, each counted with its index — the number of independent downhill directions — are constrained from below by the topology of the space they live on. The holes demand the passes. Deformation relocates a pass, reshapes it, slides it along the ridge; what deformation cannot do is delete it, because the demand does not come from the function — it comes from the space.

Say the proof structure plainly, because it is unlike the last chapter's and the contrast is the lesson. The extremal structure proves a candidate best by varying it. The obstruction structure proves a thing *exists* without ever constructing it: read the topology, count what it demands, and conclude that the landscape must contain a pass — somewhere, in some form, beyond the reach of any deformation to remove. It is proof by inevitability. The quantity the argument rests on is an invariant — a number no continuous rearrangement can change — and the conclusion is not "here is the pass" but "there is no world without one." Where the first variation's signature was one arbitrary wiggle carrying a universal quantifier, the second face's signature is an invariant carrying an existential one.

> INTUITION OFFERED, NOT CLAIM MADE. *The feeling for it: a bump under a carpet. Press it flat here and it rises there; chase it around the room all day and the one thing that never changes is that there is a bump. The carpet's excess is not a property of any place — it is a property of the carpet — and the chasing is how you discover that fact, not how you escape it.*

An illustration, not an identity — and this volume's first of two device appearances. The reader of this series has already met, in the second volume's curvature chapter, the apparatus whose self-describing loop returns changed: a residue that survives, reproducibly, however the descent is refined. That residue is offered here, again and under the same rules, as an illustration of shape *— this chapter's shape: a quantity that deformation relocates but cannot delete, the way a demanded pass slides along a ridge but never leaves the landscape. It is the same residue, worn now as obstruction-shape rather than loop-shape, and the same calibration binds: the device does not thereby do Morse theory, no invariant of any space has been computed by it, and whether its residue could ever be* earned *as an obstruction in the cited sense stays on the far side of the series' standing fence. Illustration, not identity.*

What the obstruction face cannot say is how much the surviving thing weighs. Topology demands the pass but not its height; the invariant forces existence and is silent about magnitude. When a quantity is not merely forced to exist but forced to *cancel exactly* — when two tellings of one till agree to the last coin, and the zero is a theorem rather than a coincidence — the proof at work has a different structure again, and it is the oldest one in the ledger books. That is algebra's face, and the next chapter turns to it.

Chapter 3

The Cancellation

> INTUITION OFFERED, NOT CLAIM MADE. *A bookkeeper closes the month with two ledgers kept two ways — one by account, one by day — and subtracts. The zero that comes back is not a relief; it is information. Scattered errors do not conspire to cancel to the coin: an exact zero says the two tellings were tellings of one till. The feeling to carry: an exact cancellation is never an accident, and a proof can be built out of exactly that never.*

Algebra's face of proof is the oldest and the most familiar, so familiar the reader may never have noticed it is a structure at all: the identity. Two expressions, transformed step by lawful step into one another; a sum whose neighbors annihilate in pairs until only the ends remain — the schoolroom's telescoping sum; the equality that holds not approximately, not at last, but exactly and at every instant. The legend of the structure is a schoolboy: Gauss, set with his class to sum the numbers from one to a hundred, and answering — by the account that has come down from his biographer — almost at once, having seen that the sum pairs with itself: first with last, second with second-to-last, fifty pairs of a hundred and one. The story's lesson is the face's lesson. The child did not compute harder; he found the *arrangement* in which the answer is visible — and an identity is precisely that: a rearrangement under which content shows itself unchanged.

The structure's power turns destructive — in the honest sense — when the exactness is used to prove impossibility. Take the standard specimen, published as a puzzle by Max Black in 1946: a chessboard with two opposite corners removed, and a supply of dominoes each covering two adjacent squares. Can the mutilated board be tiled? Count colors: every domino, wherever it lies, covers one dark square and one light; the two removed corners were the same color; the remaining board has thirty-two of one shade and thirty of the other. No arrangement of dominoes — none, ever, however ingenious — can cover unequal numbers with pieces that pair them. The proof visits no arrangements. It exhibits a quantity that every allowed placement preserves exactly — the color balance — and observes that the goal state disagrees with the start. Where the last chapter's invariant forced existence, this face's exactness forces *impossibility*: the ledger cannot be made to balance, because every legal entry is balanced and the books were opened unbalanced.

> INTUITION OFFERED, NOT CLAIM MADE. *The feeling for it: pairing as seeing. The child Gauss did not add; he folded the sum in half and watched the terms marry. Every proof on this face has that flavor — find the pairing, the coloring, the two tellings, and let the exactness do the work the labor would have done.*

The face's summit is a theorem that joins this chapter to the first, and it was proved by the person Hilbert had to fight his own university to hire. Emmy Noether came to Göttingen in 1915, invited by Hilbert and Klein to untangle a puzzle at the heart of the new general relativity — the status of energy conservation — while the faculty senate refused her the right to lecture on the grounds of her sex, and Hilbert made the reply that has outlived every man who voted: they were a university, not a bathhouse. Her habilitation came in 1919; the theorem this chapter needs came the year before, in the 1918 paper *Invariante Variationsprobleme*, and it says: every continuous symmetry of a variational functional forces a conserved quantity. If the functional of Chapter 1 is unchanged by shifts of time, a quantity — the one the trade calls energy — is exactly conserved along every extremal; unchanged by shifts of place, momentum; by rotations, angular momentum. The reader should see both faces meet in it: the symmetry is a variational fact — the first variation of the functional along the symmetry's direction vanishes identically — and the conservation is the exact cancellation that fact forces in the bookkeeping of every motion. The zero of energy accounting is not a fortunate habit of nature's ledgers; it is an identity, with a date and an author, riding on a symmetry. Conservation is cancellation with a reason, and the reason is Chapter 1's.

And so the fence, one paragraph as the law requires. What an exact cancellation proves is that two tellings shared content — that the functional really had the symmetry, that the coloring really was preserved, that the two ledgers really described one till. It does not prove that the world audits itself, prefers balance, or keeps books; the bookkeeper is the prover, and the zero certifies the *descriptions'* agreement, always. The reader of this series has watched this face work an honest shift already — the first volume's null method, two spellings of one content subtracted to a zero with the remainder read as signal, is this chapter's structure practiced as an instrument — and that one clause is all this chapter will say of it. Exactness belongs to the algebra. What the algebra proves is as good as its tellings, and the label on the feeling — *the zero is information* — must never harden into *the universe balances its accounts*.

What the cancellation cannot do is reach an answer that no finite rearrangement lands. Some quantities are not equal to any telling's closed form — they can only be approached, cornered, pinned between tellings that agree ever more closely and never coincide. Proving things about what you can corner but not catch is the fourth face, the numerical one, and the reader of this series will recognize its gait before the next chapter names it: it is the face the whole first volume walked on.

Chapter 4

The Squeeze

> INTUITION OFFERED, NOT CLAIM MADE. *Two hands closing on a fly. You never touch it — the fly is faster than the hands — and yet there is a moment when you know where it is, because the gap that must contain it has shrunk to almost nothing. The feeling to carry into this chapter:* caught *means* bracketed*, never* touched*, and an entire face of mathematics stands on the difference.*

The ur-instance is the oldest calculation in this series' ledgers, and it deserves its story. Archimedes, wanting the ratio of a circle's boundary to its width, did not compute it — no finite arithmetic lands it — and did not estimate it either, in the loose sense of that word. He trapped it. Inscribe a hexagon: its perimeter falls short of the circle's, and honestly so, being made of chords. Circumscribe a hexagon: too long, honestly, being made of tangents. Now double — twelve sides, twenty-four, forty-eight, ninety-six — each doubling computable by nothing but square roots, each pair of polygons a pair of sworn statements: the circle lies between us. At ninety-six sides he laid down the pen and published the two fences standing, in the *Measurement of a Circle*: the ratio exceeds three and ten seventy-firsts, and falls short of three and one seventh. The number itself was never touched, never named, never needed. What was published was a bracket and the method that tightens it at will — and that publication, twenty-two centuries old, is the complete pattern of the face this chapter teaches.

What the pattern needed, and waited two millennia for, was the license to speak of the quarry at all. Cauchy, in the *Cours d'analyse* of 1821, gave convergence its modern grammar — including the criterion that lets one certify a sequence closes in on *something* without ever producing the something

— and the arithmetization that followed, from Bolzano's early rigor of 1817 through Weierstrass's lectures, made the backbone explicit: bounded monotone sequences converge; nested closed intervals with shrinking widths share exactly one point. On that backbone the squeeze becomes a proof structure with a signature as clean as the first two faces'. To prove a statement about a number you cannot exhibit, build two sequences you can: one below the quarry, one above, both provably closing. Every property that holds for all the brackets passes to the prisoner. The quantifier lives in the construction — *for every* width, a bracket narrower — and the conclusion is a fact about a point no step of the proof ever touched.

The face has its own variation, and naming it earns the chapter's place in this volume. The first face varied the candidate; the second varied the landscape; the numerical face varies the *mesh*. Coarsen or refine the grid on which a quantity is computed, and watch the readings organize: Lewis Fry Richardson, in 1911, computing stresses in masonry by finite differences, saw that the error of a mesh reading falls in a lawful way with the mesh's fineness — and that two readings at two meshes therefore contain more than either alone, the lawful part of the error cancelling between them, the limit approached as if deferred — a method he and Gaunt named outright in 1927. The numericist's wiggle is the mesh; the analogue of the vanishing first variation is the stabilizing of readings under refinement; and the honest output is, always, a bracket with its width stated.

> Intuition offered, not claim made. *The surveyor's version of the feeling: you never stand on the point. You drive a stake on either side and swear only to the between — and your oath gets better not by touching the point but by driving the stakes closer, each advance a computation you can actually perform.*

An illustration, not an identity — and this volume's second device appearance, closing its dose. The first volume of this series documents an instrument that walks a bracket toward its quarry and stops *— at a count its own ground earned — publishing the two walls in proved order and displaying, never asserting, the point between them. That practice is offered here as the shape of this chapter's structure worn honestly by a machine: brackets constructed, walls certified, the point left untouched. One calibration differs from this volume's first exhibit and is stated plainly: the instrument's brackets are its own carried structure — the walls and their order are checked mechanically, as its documentation records — so what is illustrated is not a resemblance but a practice. What stays behind the fence is the* limit*: this chapter's theorems pass to a point at the end of an infinite closing, and the instrument, by its own law, never does — it stops at its earned count and asserts the interval. Whether its stopped bracket could ever be earned as this chapter's convergence is a question the series leaves exactly where it leaves all such questions: owed, not crossed. Illustration — and here, honest practice — but not identity.*

Take stock, because the volume's tour of faces is complete. To prove best: vary the candidate, one arbitrary wiggle carrying every rival — the extremal structure. To prove must-exist: read the invariant the space imposes — the obstruction structure. To prove exactly: find the arrangement in which the content cancels to zero — the identity structure. To prove about the unreachable: trap it between computable fences that close — the convergence structure. Four faces, four signatures, every one cited, every picture labeled. What remains is the suspicion the reader has surely formed by now — that these are not four subjects but four lights on one — and the next chapter takes one proof the reader already owns and turns it under all four, to see.

Chapter 5

The Four as One

> intuition offered, not claim made. *Hold one object up to four lamps and it throws four shadows, no two alike; a stranger shown only the shadows might catalog four objects. The feeling this chapter tests: the reader has just cataloged four proof structures — and they may be shadows. Not four subjects; four lights on one.*

The test must be concrete or it is only the feeling again, so take the one theorem this series has owned the longest — the shortest path, the geodesic of the second volume — and turn it under each lamp in order.

Under the first lamp, the geodesic is Chapter 1's original citizen. The claim quantifies over every rival path; the proof adds one arbitrary wiggle, differentiates, and demands the first-order change vanish; the Euler–Lagrange condensation turns that demand into the pointwise equation the second volume's ruler-selected carrying rule already satisfies. Nothing new here — this was the chapter's own closing example — except the noticing: the geodesic is, first of all, an *extremal* theorem.

Under the second lamp, ask a different question: not *which* path is shortest, but whether a shortest path of a given kind must exist at all. On the doughnut of Chapter 2, consider loops that go around the hole. No such loop can be shrunk to a point — that is the surface's shape speaking — and among them one may hunt the shortest. The hunt succeeds, and its trophy is a closed geodesic: the loop that cannot be removed can be tightened until it is extremal, and the tightening cannot lose the loop, because the hole demands it. The argument descends from Birkhoff's 1917 study of closed geodesics, and the reader should notice where they have seen the machinery before: Morse's 1934 calculus of variations in the large — the second chapter's summit — was built with exactly this example as its engine. The geodesic is, secondly, an *obstruction* theorem: some geodesics exist because the space leaves no world without them.

Under the third lamp, let the ground have a symmetry. On a surface of revolution — a vase, a globe — spin is a symmetry of the whole variational setup, and Chapter 3's summit then promises a conserved quantity along every geodesic. The promise is kept by a relation the texts teach under Clairaut's name, three centuries old and standard in the geometry references this series already cites: a definite combination of the path's distance from the axis and its bearing stays *exactly* constant along the geodesic, from launch to landing. A navigator can check it at any two instants and the books balance to the coin. The geodesic is, thirdly, a *cancellation* theorem: its symmetry forces an identity, and the identity rides the whole path.

Under the fourth lamp, try to hold an actual geodesic in the hand — the shortest route between two real cities on the real globe — and the face of Chapter 4 takes over, because no finite arithmetic lands it. Chain straight segments between waypoints: every chained path is honestly too long or honestly measurable, and refining the waypoints tightens the estimate lawfully — the mesh is the wiggle, as the fourth chapter taught, and the honest output is a bracket: the true shortest length pinned between computable fences that close under refinement. The geodesic is, fourthly, a *squeeze* theorem: approachable to any width, touchable never.

One theorem; four proofs; four different things *proved.* The lamps do not repeat one another — the extremal light shows which path, the obstruction light shows that a path must exist, the cancellation light shows what rides the path unchanged, the squeeze light shows how the path is caught by finite hands — and yet every one of them is the same gesture performed in a different quantity: *vary, and watch.* Vary the path, watch the first order. Vary the landscape, watch the invariant. Vary the telling, watch the cancellation. Vary the mesh, watch the bracket. That gesture is what this volume has meant by a variational proof structure, and the four faces are its four disciplined forms.

> INTUITION OFFERED, NOT CLAIM MADE. *The series has an image for a thing known only through its faces, and this is the one paragraph where the image is named. A cube casts square shadows; a four-dimensional cube — the tesseract of the older volumes — casts* cubic *ones, and an inhabitant of three dimensions learns it only shadow by*

> *shadow. Four proof structures, one gesture: the reader may hold the image. It is an image.*

And the fence, brief because the label above already carries the weight. That the four faces are four forms of one gesture is a *reading* — this book's arrangement of cited material, the kind of claim its own preface licensed and no stronger. It is not a theorem of metaphysics; no result named in this chapter asserts that all mathematics is variation, and the reader who leaves believing that has taken more than was sold. What was sold: one worked theorem, four cited proofs, one gesture visible in all four. The shadows agree; the object is an image; the citations are the content.

The tour would end here if proofs were all the world asked mathematics to do. But the reader of a series about a measuring instrument knows the other thing the world asks, and the four clean faces share an assumption worth noticing before the next chapter breaks it: every one of them conserves something — the extremal its stationarity, the invariant its count, the identity its balance, the bracket its quarry. Nothing in the four ever *leaks.* The next chapter is about the leak — the systems that forget, the friction that variational cleanliness cannot fully house — and about why the readings nobody understands well tend to live exactly there.

Chapter 6

The Leak

The last chapter closed on a noticing: all four faces conserve. The extremal keeps its stationarity, the invariant its count, the identity its balance, the bracket its quarry — nothing leaks. So bring the tour to an honest bench and let something leak. A pendulum in air: Chapter 1 derives its swing beautifully — and the swing dies. Each arc is lower than the last; the energy Chapter 3's summit promised to conserve drains away into the air's warmth; and after enough of that, the pendulum hangs at rest with no trace of how it started. The system has *forgotten*. Where does the variational tour stand before a thing like that? The honest answer is: awkwardly — and this chapter is about the awkwardness, because naming where a method stops is a duty this series owes its readers. The last volume named where the no-ruler construction stopped, at the metric; this volume names where the clean variational structures stop, at the leak.

The awkwardness is precise, not vague, and it has a classical half-remedy with a name and a date. Friction of the common kind — resistance proportional to how fast you push through it — does not descend from any potential, so it cannot be folded into the functional whose variation Chapter 1 taught; the clean Euler–Lagrange house has no room for it. Rayleigh, in the 1870s, in the work gathered into *The Theory of Sound*, built the standard annex: a second scalar — the dissipation function — kept alongside the functional, whose gradient enters the equations of motion as a correction. It works; the trade uses it daily; and the reader should see it for what it is architecturally: an *annex*, not a room. The dissipative term rides beside the variational structure, not inside it. And the deeper faces lose more than convenience. Noether's bargain — Chapter 3's summit — trades a symmetry of the functional for an exact conservation; the damped pendulum keeps the symmetry of its laws under time-shift and conserves nothing, because the bargain's premise — that the dynamics descend whole from a functional — is exactly what the leak breaks. Of the four faces, only the fourth keeps its full footing: brackets do not care whether the system remembers, and the numericist integrates damped systems every working day. The clean faces leak; the squeeze survives; and that asymmetry is the boundary's shape.

> INTUITION OFFERED, NOT CLAIM MADE. *Now the feeling this chapter exists to hand over, at the operator's own prompting, and the label binds every sentence of it. A system with a leak* forgets. *Whatever the start — however violent the launch, however particular the history — the transients ring down, the details drain into the annex, and what remains is the settled thing: the rest state, the steady hum, the long-run rhythm that no longer answers questions about the beginning. Now consider what an instrument sees when it reads such a system. A reading taken after the forgetting reflects the settled thing only; the itemized bookkeeping the four clean faces promise — this much from the launch, this much conserved, this much exactly cancelled — is precisely what the leak has destroyed. The intuition on offer: when a reading resists the clean accountings — when it will not itemize, will not conserve, will not cancel to the coin — it is* worth asking *whether one is reading a system mid-forgetting, and the leak is where such readings would naturally pool. That is a place to look, handed to the reader as a lens. It is not a finding.*

Where is the leak itself well understood? The trade's canonical home for viscosity — the word for the leak in a flowing medium — is the dissipative term of the Navier–Stokes equations, and the reader of this series knows that address already: it is the fluid face, promised in the founding text of the series' apparatus, named as owed in every volume since, and paid in none — including, emphatically, this one. This chapter borrows the *word* viscosity for its lens and returns the word unspent: no fluid is constructed here, no dissipative dynamics derived, and the fourth station stands exactly as owed as it stood a chapter ago.

The fence, first and last, because this chapter is the volume's nearest approach to its own failure mode. Everything above the citations in this chapter is intuition, and labeled; the chapter asserts, as mathematics, only what Rayleigh's annex and Noether's premise already contain. The phrase the reader may have heard in this series' wider conversation — the viscosity of space-time — is an image*, of the kind the preface licensed: a way of holding the lens, not a claim about the world, not a property this series has measured, not physics any volume of it has earned. And the lens itself obeys the lens's law: that ill-understood readings* might *pool at the leak is a place to look, offered as such; whether any particular reading is in fact dissipative must be earned per-reading, off that reading's own construction, and no reading has been so earned here. The boundary of the variational method is real, cited, and this chapter's content. The territory past the boundary is owed — to the volume that pays the fluid its debt — and this book stops at the boundary line, on purpose, the way its predecessor stopped at a door.*

What remains for this volume is only its accounting: the citation ledger, chapter by chapter; the intuition ledger the preface promised — every picture this book handed over, listed beside the citation it re-sees; and the honest grades of the faces used. That is the back matter's business, and the final chapter's.

Chapter 7

The Ledgers

This book made two promises on its first page, and the audit of a two-law book has two ledgers. Every claim traced to a name and a date: the citation ledger, as in the volumes before. And — new to this volume, because its licence was new — every picture listed beside the citation it re-sees: the intuition ledger, which exists so that the preface's law is checkable line by line. If any labeled intuition in this book asserts one thing its listed neighbor does not already contain, that row is a defect; the book asks to be audited against exactly that standard.

The citation ledger

Preface

- Legendre, 1805 — the method of least squares, published first; Gauss, 1809 and 1821 — the private prior use and the theory, for the overdetermined systems of astronomy and geodesy. Back: the honest job of many.
- The trilateration arithmetic of the previous volume's preface, carried forward as the earn's ground.

Chapter 1 — The First Variation

- Johann Bernoulli, 1696 — the brachistochrone challenge; the Newton solution overnight, by the account descended through his niece; the cycloid; the five solvers.

- Fermat, 1662 — least time, the anticipation.
- Euler, 1744 — the treatise and the equation; Lagrange, 1755 and 1788 — the variation itself and the *Mécanique analytique*, with Euler's adoption and naming told as the attribution it is.

Chapter 2 — The Second Variation

- Legendre, 1786 — the sign condition.
- Jacobi, 1837 — conjugate points.
- Morse, 1934 (*The Calculus of Variations in the Large*) — critical points bounded below by topology; the index.

Chapter 3 — The Cancellation

- The schoolboy Gauss pairing, by his biographer's account — carried as story, flagged as such in the telling.
- Black, 1946 — the mutilated chessboard.
- Noether, 1918 (*Invariante Variationsprobleme*) — symmetry forces conservation; the Göttingen story with Hilbert's reply.

Chapter 4 — The Squeeze

- Archimedes (*Measurement of a Circle*) — the polygon squeeze, the two fences standing at ninety-six sides.
- Bolzano, 1817; Cauchy, 1821; Weierstrass — the rigor backbone (convergence certified without the limit; nested intervals).
- Richardson, 1911; Richardson and Gaunt, 1927 — the mesh as the face's variation; the deferred approach to the limit.

Chapter 5 — The Four as One

- Birkhoff, 1917 — closed geodesics; with Morse, 1934, whose engine was this example.
- Clairaut's relation — via the standard geometry references this series cites (do Carmo).
- The face-chapters' own citations, re-spent where each lamp was lit.

Chapter 6 — The Leak

- Rayleigh, the 1870s (*The Theory of Sound*) — the dissipation function, the annex beside the variational house.
- The Navier–Stokes dissipative term — named as the leak's canonical address and as the series' owed face; cited as owed, spent nowhere.

The intuition ledger

Eleven labeled intuitions were handed over; each is listed with the cited material it re-sees, and none adds a claim its neighbor lacks.

- I1. The soap film, the chain, the ray (Ch. 1) — re-sees Fermat's least time and the functionals of Euler and Lagrange.
- I2. Stand-and-jiggle (Ch. 1) — re-sees Lagrange's variation: one arbitrary perturbation carrying all rivals.
- I3. The mountain pass (Ch. 2) — re-sees Legendre's condition and Jacobi's classification: flatness at first order, truth at second.
- I4. The bump under the carpet (Ch. 2) — re-sees Morse's invariance: what deformation relocates but cannot delete.
- I5. The bookkeeper's zero (Ch. 3) — re-sees the exact identity: an exact cancellation certifies shared content.
- I6. Pairing as seeing (Ch. 3) — re-sees the rearrangement identity of the Gauss story.

I7. The fly between the hands (Ch. 4) — re-sees the squeeze: caught means bracketed.

I8. The surveyor's stakes (Ch. 4) — re-sees bracket construction: swearing only to the between.

I9. Four lamps, four shadows (Ch. 5) — re-sees the four cited structures under one gesture; the frame of the worked example.

I10. The tesseract (Ch. 5) — the series' image for shadow-by-shadow knowledge; labeled "It is an image," and it is listed here as exactly that: an image, with no theorem beside it because it asserts none.

I11. The forgetting (Ch. 6) — re-sees dissipation's ring-down (Rayleigh's annex); offered as a place to look; "not a finding" recorded here as its standing grade.

The face-grade ledger

This volume used the four faces of mathematics as proof structures, and each stands anchored to a cited summit: the extremal face on Euler–Lagrange; the obstruction face on Morse; the cancellation face on Noether; the convergence face on the line from Archimedes to Weierstrass. Its two device appearances are graded as their exhibits stated: the second chapter's, a shape-resemblance with the obstruction reading owed; the fourth chapter's, a carried practice — the bracketing — with the limit owed. The boundary of the method was named in the sixth chapter and is a grade in itself: the three clean faces leak at dissipation, the fourth survives, and the territory past the boundary belongs to the volume that pays the fluid its debt — the face that remains, here as in every volume of this series, owed. The unity of the four is graded exactly as sold: a reading.

The closing line

A book of intuitions succeeds when its reader can no longer tell, from the inside, which understanding arrived by proof and which by picture — and its honesty consists in the ledgers that can always tell. The mathematics belongs to the names above; the pictures re-see it and add nothing; the arrangement is this book's only claim, and the arrangement is a reading plan, as it was in the volume before. The four faces are now felt as well as checked. What waits past the boundary line — the leak's own volume, the fluid's unpaid station — waits with its debt recorded in three ledgers now. The feeling is yours; the receipts are here.